



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 7: $v^2 - u^2 = 2as$

One equation that links speeds and distance, with no time in it at all.

Background Knowledge

Read this as you come in. Highlight the key words, and use the space to add your own notes.

A. Distance, speed and velocity

	Scalar — size only	Vector — size and direction
How far	distance , in metres (m)	displacement : how far from the start, and which way
How fast	speed , in metres per second (m/s)	velocity : speed in a stated direction

$$s = vt \quad \text{distance travelled} = \text{speed} \times \text{time}$$

A car at 20 m/s for 5.0 s travels $20 \times 5.0 = 100$ m. This works when the speed is **steady**. When the speed changes, use the **average** speed.

B. Acceleration

Acceleration is the **rate of change of velocity**: how much the velocity changes each second.

$$a = \frac{\Delta v}{t}$$

acceleration = change in velocity $\div$ time

$$\Delta v = v - u \quad (\text{final} - \text{initial})$$

- u is the **initial** velocity, v the **final** velocity. Unit of acceleration: **m/s²** (metres per second, every second).
- Speeding up: a is **positive**. Slowing down: a is **negative**. Finish minus start, in that order, and the sign comes out right by itself.
- **Uniform** acceleration means the velocity changes by the same amount every second. An object falling freely near the Earth accelerates at about **9.8 m/s²**.

A bike goes from 2.0 m/s to 8.0 m/s in 3.0 s: $a = (8.0 - 2.0) \div 3.0 = 2.0 \text{ m/s}^2$.

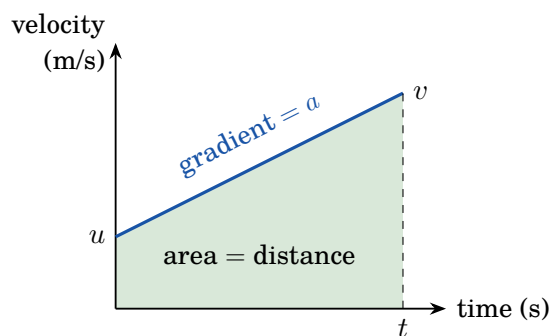
C. Reading the motion graphs

Distance–time graph

- **gradient = speed**
- a flat line: stationary
- a steeper line: faster

Velocity–time graph

- **gradient = acceleration**
- **area under the line = distance travelled**
- a flat line: constant velocity



$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

A straight sloping line on a velocity–time graph means **uniform** acceleration. That is the case today's equation is built for.

D. How this ties together — and where today goes

Everything so far has **time** in it: $s = vt$, and $a = \Delta v \div t$.

Plenty of real questions give you **no time at all**. A car brakes from 20 m/s to a stop: how far does it skid? An aeroplane needs to reach 60 m/s: how long must the runway be? You know the speeds and the acceleration, and you want the distance.

The velocity–time graph in C already holds the answer. The shaded area is the distance, and it equals the **average velocity** $\times$ **time**:

$$s = \frac{u + v}{2} \times t \quad \text{and from B,} \quad t = \frac{v - u}{a}$$

Put the second into the first and the time cancels out. What is left is one equation linking u , v , a and s , with no t in it:

$$v^2 - u^2 = 2as$$

Today's question: if you know how fast something starts, how fast it finishes and how hard it accelerates, how far does it travel?

The equation — it is on your equation sheet

$$v^2 - u^2 = 2 a s$$

$$(\text{final velocity})^2 - (\text{initial velocity})^2 = 2 \times \text{acceleration} \times \text{distance}$$

Symbol	Quantity	Unit	Symbol	Quantity	Unit
u	initial velocity	m/s	a	acceleration	m/s ²
v	final velocity	m/s	s	distance	m

When to use it: the acceleration is **uniform** (steady), and the question gives you three of u , v , a , s and asks for the fourth. There is no time in it, so it is the one to reach for when no time is given.

Reading the letters: a u in the equation tells you v means the **final** velocity. **From rest** means $u = 0$. **Comes to a stop** means $v = 0$. A slowing object has a **negative** acceleration.

The four arrangements — pick the one that has your unknown on its own

Find s

$$s = \frac{v^2 - u^2}{2a}$$

Find a

$$a = \frac{v^2 - u^2}{2s}$$

Find v

$$v = \sqrt{u^2 + 2as}$$

Find u

$$u = \sqrt{v^2 - 2as}$$

Square first, then subtract or add. For v or u , the square root is the **last** step.

Worked example 1 — find the distance

A cyclist speeds up from 3.0 m/s to 7.0 m/s with an acceleration of 0.50 m/s². Calculate the distance travelled while accelerating.

- List** $u = 3.0$, $v = 7.0$, $a = 0.50$, $s = ?$
- Square first** $v^2 = 49$, $u^2 = 9.0$
- Substitute** $s = \frac{49 - 9.0}{2 \times 0.50} = \frac{40}{1.0}$
- Answer** $s = 40 \text{ m}$

Worked example 2 — find the final velocity

A trolley moving at 5.0 m/s accelerates at 2.0 m/s² over a distance of 30 m. Calculate its final velocity. Give your answer to 2 significant figures.

- List** $u = 5.0$, $a = 2.0$, $s = 30$, $v = ?$
- Find v^2** $v^2 = u^2 + 2as = 25 + (2 \times 2.0 \times 30) = 25 + 120 = 145$
- Square root last** $v = \sqrt{145} = 12.04 \dots$
- Answer** $v = 12 \text{ m/s}$ (2 s.f.)

The check. Put your answer back into $v^2 - u^2 = 2as$ and work out **each side on its own**. The two sides give the **same number**. In worked example 1: $v^2 - u^2 = 49 - 9 = 40$ and $2as = 2 \times 0.50 \times 40 = 40$. They match, so $s = 40 \text{ m}$ is right.

Questions

Write down what you are given, then the equation, the substitution and the answer with its unit. Use the check on at least one answer in each section.

Everyone — find the distance

- 1) **Calculate** the distance a runner covers while accelerating from rest to 8.0 m/s at 2.0 m/s². [3]

$$u = 0, \quad s = \frac{v^2 - u^2}{2a} = \frac{64 - 0}{2 \times 2.0} = \frac{64}{4.0} = \mathbf{16 \text{ m}}$$

- 2) **Calculate** the distance a car travels while it speeds up from 6.0 m/s to 14 m/s with an acceleration of 1.6 m/s². [3]

$$s = \frac{196 - 36}{2 \times 1.6} = \frac{160}{3.2} = \mathbf{50 \text{ m}}$$

- 3) **Calculate** the distance a skateboarder rolls downhill while speeding up from 1.0 m/s to 5.0 m/s with an acceleration of 0.80 m/s². [3]

$$s = \frac{25 - 1.0}{2 \times 0.80} = \frac{24}{1.6} = \mathbf{15 \text{ m}}$$

Most — rearrange for the unknown

- 4) **Calculate** the braking distance of a car that slows from 20 m/s to a stop with an acceleration of -5.0 m/s². A distance comes out positive. [3]

$$v = 0, \quad a = -5.0, \quad s = \frac{0 - 400}{2 \times (-5.0)} = \frac{-400}{-10} = \mathbf{40 \text{ m}}$$

- 5) **Calculate** the acceleration of an aeroplane that starts from rest and reaches 60 m/s along a runway 900 m long. [3]

$$a = \frac{v^2 - u^2}{2s} = \frac{3600 - 0}{2 \times 900} = \frac{3600}{1800} = \mathbf{2.0 \text{ m/s}^2}$$

- 6) **Calculate** the final velocity of a ball that is rolling at 3.0 m/s and then accelerates at 1.5 m/s² over 12 m. Give your answer to 2 significant figures. [3]

$$v^2 = u^2 + 2as = 9.0 + (2 \times 1.5 \times 12) = 9.0 + 36 = 45, \quad v = \sqrt{45} = 6.708 \dots = \mathbf{6.7 \text{ m/s}}$$

Stretch — two equations in one question

- 7) A stone is dropped from rest from a bridge and falls 20 m to the water. It falls with an acceleration of 9.8 m/s².

- (a) Calculate the speed of the stone as it reaches the water. Give your answer to 2 significant figures.

$$v^2 = 0 + (2 \times 9.8 \times 20) = 392, \quad v = \sqrt{392} = 19.80 \dots = \mathbf{20 \text{ m/s (2 s.f.)}}$$

[3]

- (b) Use your answer to (a) and $a = \frac{\Delta v}{t}$ to calculate how long the stone takes to fall.

$$t = \frac{\Delta v}{a} = \frac{19.8 - 0}{9.8} = 2.02 \dots = \mathbf{2.0 \text{ s}}$$

[2]

8) A train speeds up steadily from 8.0 m/s to 16 m/s over a distance of 240 m.

(a) Calculate the acceleration of the train.

$$a = \frac{256 - 64}{2 \times 240} = \frac{192}{480} = \mathbf{0.40 \text{ m/s}^2}$$

[3]

(b) Calculate the time the train takes to cover the 240 m.

$$t = \frac{\Delta v}{a} = \frac{16 - 8.0}{0.40} = \mathbf{20 \text{ s}}$$

[2]

(c) Calculate the average speed of the train over the 240 m. Your answer sits between its initial and final velocities.

$$\frac{240}{20} = \mathbf{12 \text{ m/s}} \text{ — between 8.0 and 16, and exactly halfway, as it is for uniform acceleration.}$$

[1]